Project Development Phase – Performance Test

Global Olympic Sports Performance Analysis (Tableau Desktop Suite)

Date: 10 February 2026	**Team ID:** PNT2026TMID092418
Project Name: Global Olympic Sports Performance Analysis	**Maximum Marks:** 5 Marks

Tableau Model Performance Testing Parameters

Performance test evaluation measuring data engine rendering, extract preparation, filter query latency, calculated field metrics, dashboard visual layout, and story storytelling presentation performance.

S.No.	Parameter	Screenshot / Values & Technical Details
1	**Data Rendered**	**Data Extract Connection:** Tableau Hyper Extract (`Olympic_Performance.hyper`) connected to Snowflake Data Warehouse. • **Total Records Rendered:** 38,574 historical medal records & 271,116 athlete entries. • **Extract Load Time: 420 ms** initial cold load, sub-50 ms for cached memory queries.
2	**Data Preprocessing**	**Tableau Prep Builder Flow:** • Cleaned country string variations into uniform 60 delegation profiles. • Joined `athletes` and `medal_results` using inner join on `athlete_id`. • Pivoted medal type fields (Gold, Silver, Bronze) for dynamic visual aggregation. • Validated zero null values across core visual analytical dimensions.
3	**Utilization of Filters**	**Global Interactive Quick-Filters Applied:** • *Country Filter:* Drop-down selector across 60 participating nations. • *Sport Category Filter:* Multi-select list across 50 sports categories. • *Olympic Year & Season Filter:* Range slider (1992–2026) and radio toggles (Summer / Winter). • *Filter Performance Test:* Dashboard rendering response time = **1.1 seconds** across complex parameter switches.
4	**Calculation Fields Used**	**Custom Tableau Calculated Fields:** • [Avg Medals per Athlete]: SUM([Total Medals]) / COUNTD([Athlete ID]) (Rowing: 6.916, Shooting: 6.731) • [Medal Efficiency %]: (SUM([Total Medals]) / SUM([Sports Investment USD])) * 1000000 • [Host Advantage Flag]: IF [Delegation Country] = [Host Country] THEN 'Host' ELSE 'Visitor' END • [Regional Efficiency Leader]: South America at 49.64%.
5	**Dashboard Design (No of Visualizations / Graphs)**	**Total Visualizations across 2 Interactive Dashboards: 8 Visual Charts** • **Dashboard 1 (Global Performance):** Horizontal Bar Chart (Avg Medals per Athlete), Ranked Country Column Chart (USA 735, China 696), Trend Line Chart (1992–2026: 305 to 363 to 334). • **Dashboard 2 (Regional Efficiency):** Investment vs. Medal Bubble Chart (Slovakia, Ethiopia), Regional Efficiency Donut Chart (South America 49.64%), Host Nation Multi-Bar Comparison Chart (Japan 105).
6	**Story Design (No of Visualizations / Graphs)**	**Tableau Story Narrative Suite: 4 Interactive Story Points** • *Point 1:* Per-Athlete Medal Return & High-Yield Sports Analysis (Rowing/Shooting). • *Point 2:* Global Medal Powerhouses Benchmark (USA & China). • *Point 3:* Capital Efficiency & High-ROI Nations (South America 49.64%). • *Point 4:* Host Nation Advantage (Japan 105) & 34-Year Olympic Expansion.